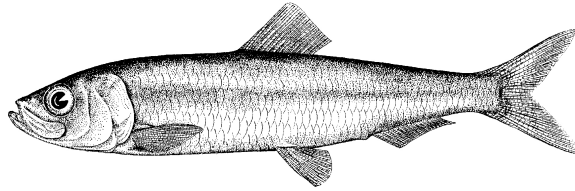


Pacific Herring preliminary data summary for Haida Gwaii 2026

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Pacific Herring (*Clupea pallasii*). Image credit: [Fisheries and Oceans Canada](#).

Disclaimer This report contains preliminary data collected for Pacific Herring in 2026 in the Haida Gwaii major stock assessment region (SAR). These data may differ from data used and presented in the final stock assessment.

1 Context

Pacific Herring (*Clupea pallasii*) in British Columbia are assessed as 5 major and 2 minor stock assessment regions (SARs), and data are collected and summarized on this scale (Table 1, Figure 1). Note that formal stock assessments are only done for major SARs. The Pacific Herring data collection program includes fishery-dependent and -independent data from 1951 to 2026. This includes annual time series of commercial catch data, biological samples (providing information on proportion-at-age and weight-at-age), and spawn index data conducted using a combination of surface and SCUBA surveys. In some areas, industry- and/or First Nations-operated in-season soundings programs are also conducted, and this information is used by resource managers, First Nations, and

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stakeholders to locate fish and identify areas of high and low Pacific Herring biomass to plan harvesting activities. In-season acoustic soundings are not used by stock assessment to inform the estimation of spawning biomass.

The following is a description of data collected for Pacific Herring in 2026 in the Haida Gwaii major SAR (Figure 2). Data collected outside the SAR boundary are not included in this summary, and are not used for the purposes of stock assessment. Although we summarise data at the scale of the SAR for stock assessments, we summarise data at finer spatial scales in this report: Locations are nested within Sections, Sections are nested within Statistical Areas, and Statistical Areas are nested within SARs (Table 2). Note that we refer to ‘year’ instead of ‘herring season’ in this report; therefore 2026 refers to the 2025/2026 Pacific Herring season.

2 Data collection programs

Biological samples were collected by the *Queens Reach*, a seine test charter vessel funded by DFO. The primary purpose of the test charter vessel was to collect biological samples from main bodies of herring from Haida Gwaii major (priority) and Area 2W minor stock areas, identified from soundings.

- The *Queens Reach* operated a 21 day charter from March 12th to April 1st, collecting samples from HG and Area 2 West.
- The *Haida Spirit* operated an 15 day dive charter from April 4th to April 18th.
- The *Ten Forward* operated a spawn reconnaissance charter for 19 days from March 27th to April 15th with surface survey work.

3 Catch and biological samples

In the 1950s and 1960s, the reduction fishery dominated Pacific Herring catch; starting in the 1970s, catch has been predominantly from roe seine and gillnet fisheries. The reduction fishery is different from current fisheries in several ways. First, the reduction fishery caught Pacific Herring of all ages, whereas current fisheries target spawning (i.e., mature) fish. Thus, reduction fisheries included an unknown number of age-1 fish which are not typically caught in current fisheries. Second, the reduction fishery has some uncertainty regarding the quantity and location of catch; however catches have been resolved to SAR and Statistical Area using fish slips as best as possible. For the roe gillnet fishery, all Pacific Herring catch has been validated by a dockside monitoring program since 1998; the catch validation program started in 1999 for the roe seine fishery. Finally, the reduction fishery operated during the summer and winter months, whereas roe fisheries typically target spawning fish between February and April.

Landed commercial catch of Pacific Herring by year and fishery is shown in Table 3 and Figure 3. Total harvested spawn-on-kelp (SOK) in 2026 in the Haida Gwaii major SAR is shown in Table 4; we also calculate the estimated spawning biomass associated

with SOK harvest. See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass.

In 2026, 4 Pacific Herring biological samples were collected and processed for the Haida Gwaii major SAR (Table 5, Table 6). The locations in which the biological samples were collected are presented in Figure 4. Included herein are biological summaries of observed proportion-, number-, weight-, and length-at-age (Figure 5, Table 7, and Figure 6, respectively). We also show the percent change in weight and length for age-3 and age-6 fish (Figure 7 & Figure 8, respectively). Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. Only representative biological samples are included, where ‘representative’ indicates whether the Pacific Herring sample in the set accurately reflects the larger Pacific Herring school.

4 Spawn survey data

Pacific Herring spawn surveys were conducted at 29 individual locations in 2026 in the Haida Gwaii major SAR (Table 8, and Figure 9). A summary of spawn from the last decade (2016 to 2025) is shown in Figure 10. Figure 11 shows spawn start date by decade and Statistical Area. Spawn surveys are conducted to estimate the spawn length, width, number of egg layers, and substrate type, and these data are used to estimate the index of spawning biomass (i.e., the spawn index; Figure 12, Figure 13, Figure 14, Figure 15, Table 9, and Figure 16). See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Therefore, these data do not represent model estimates of spawning biomass, and are considered the minimum observed spawning biomass derived from egg counts. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

Some Pacific Herring Sections contribute more than others to the total spawn index, and the percentage contributed by Section varies yearly (Figure 15b, Figure 17). For example, in 2026, Section 021 contributed the most to the spawn index (66%). As with Sections, some Statistical Areas contribute more than others to the total spawn index (Figure 15c, Figure 18). An animation shows the spawn index by spawn survey location from 1951 to 2026 (Figure 19).

5 First Nations observations

The following observations were contributed by representatives of First Nations communities. These observations provide context and perspectives to this data report. In some cases we make edits for clarity and brevity, but we do not change the intent or substance of responses. Note: if your observations are missing, submit them [here](#).

5.1 Haida Nation

Surveyed the Haida Gwaii major stock assessment area from Cumshewa Inlet south to Skincuttle Inlet. In 2026 the Haida Fisheries Program (HFP) was contracting to complete the herring spawn reconnaissance charter (19 days between March 25th and April 14th) and the herring spawn dive survey (15 days between April 3rd to 17th).

Just prior to the start of the herring recon charter the HFP vessel *Haida Guardian* had a major mechanical issue which resulted in the HFP contracting Leandre Vigneault and his vessel the *Ten Forward* to assist in the charter. Leandre's experience in the recon charter assisted greatly in identifying, tracking and (when required) conducting surface surveys. HFP staff on the recon charter included Dan McNeill for the first 8 days and Ben Penna for the following 11 days. Both Dan and Ben are very experienced in herring spawn surveys on Haida Gwaii. As a result, the Haida Gwaii herring recon charter went very well. A total of 74,490 meters of spawn was identified and reported, and 6 of the smaller spawns were assessed using surface survey methodology.

The herring spawn dive survey began with an abundance of spawn in upper Burnaby Strait. Other spawns were located in inner Skincuttle Inlet, Poole Inlet and out toward Scudder Point. Spawns were heaviest along the East side of Huxley Island in Upper Burnaby. By mid charter it became apparent that all spawn transects could not be completed over the 15 charter days, so a 'skip one' pattern for survey transects was adopted on the 11th day of the surveys. Approximately 110 transects were completed over the 15 day charter.

Sounding and sampling of herring schools (prior to spawns) on Haida Gwaii was conducted by the charter vessel *Queens Reach*. As a result, HFP staff did not have opportunities to provide reliable observations related to fish behavior and characteristics. During the recon and spawn charters whales were observed as well as an abundance of sea lions and birds (similar to recent years).

Herring spawns on Haida Gwaii in 2026 appear to be similar to recent year. Generally, most of the herring spawns rated medium to good were concentrated in upper Burnaby Strait. Other spawns throughout East Moresby were mostly very light to medium. Spawns in Selwyn Inlet continue to be very minimal compared to historical levels (a trend which has been the case for the past 5+ years).

- Approximately 10,000 lbs of k'aaw was harvested from the major stock assessment area of East Moresby (primarily from the east side of Huxley Island Harbour in upper Burnaby Strait) in early to mid-April. Product was reported to be moderate to very good quality.
- Most of the traditional k'aaw harvest was conducted by three Haida individuals: one harvesting on behalf of the Skidegate Band, another harvesting on behalf of the Old Massett Band, and the third harvesting for personal use.
- There have been no known harvests of k'aaw from other locations on Haida Gwaii (e.g., Skidegate Inlet).

6 General observations

The following observations were reported by DFO Resource Management staff and DFO Science staff. These observations provide additional context to this data report.

- Once again, the main herring aggregations and majority of spawn occurred in the Upper Burnaby area while there was less spawn in Skincuttle than recent years.
- Spawns occurred in more areas of Juan Perez Sound than seen in recent years including Sedgwick Bay as well as De La Beche, Hutton and Marshall Inlets.
- Spawn timing in the Skincuttle area was approximately 10 days earlier than the previous five-year average. Spawn timing in other parts of the major stock area was similar to the previous five-year average but the duration of spawning in the Burnaby Island area was shorter than most years.
- Water temperatures in the major stock assessment area were approximately 2 degrees warmer than recent years.
- All spawns in the major assessment area were identified and dive assessments were completed for the majority. Surface surveys were completed by the reconnaissance vessel for some smaller spawns in Cumshewa Inlet, Atli Inlet, Juan Perez, Sound, Island Bay, and Poole Inlet.
- Once again in 2026 the abundance of Humpback whales in the Burnaby Strait area (more than 25) hampered test setting efforts.

7 Tables

Table 1. Pacific Herring stock assessment regions (SARs) in British Columbia.

Name	Code	Type
Haida Gwaii	HG	Major
Prince Rupert District	PRD	Major
Central Coast	CC	Major
Strait of Georgia	SoG	Major
West Coast of Vancouver Island	WCVI	Major
Area 27	A27	Minor
Area 2 West	A2W	Minor

Table 2. Statistical Areas and Sections for Pacific Herring in the Haida Gwaii major stock assessment region (SAR).

Region	Statistical Area	Section
Haida Gwaii	00	006
Haida Gwaii	02	021
Haida Gwaii	02	023
Haida Gwaii	02	024
Haida Gwaii	02	025

Table 3. Total landed commercial catch of Pacific Herring in metric tonnes (t) by gear type in 2026 in the Haida Gwaii major stock assessment region (SAR). Legend: ‘Other’ represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; ‘RoeSN’ represents the roe seine fishery; and ‘RoeGN’ represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Note: data may be withheld due to privacy concerns (WP).

Gear	Catch (t)
Other	0
RoeSN	0
RoeGN	0

Table 4. Total harvested Pacific Herring spawn-on-kelp (SOK) in pounds (lb), and the associated estimate of spawning biomass in metric tonnes (t) from 2016 to 2026 in the Haida Gwaii major stock assessment region (SAR). See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. Note: data may be withheld due to privacy concerns (WP).

Year	Harvest (lb)	Spawning biomass (t)
2016	0	0
2017	0	0
2018	0	0
2019	0	0
2020	0	0
2021	0	0
2022	0	0
2023	0	0
2024	0	0
2025	0	0
2026	0	0

Table 5. Number of Pacific Herring biological samples processed from 2016 to 2026 in the Haida Gwaii major stock assessment region (SAR). Each sample is approximately 100 fish. Note: Nearshore samples are not used in stock assessments.

Year	Number of samples			
	Commercial	Test	Nearshore	Total
2016	0	5	0	5
2017	0	8	0	8
2018	0	11	0	11
2019	0	10	0	10
2020	0	12	0	12
2021	0	11	0	11
2022	0	9	0	9
2023	0	7	0	7
2024	0	7	0	7
2025	0	8	0	8
2026	0	4	0	4

Table 6. Number and type of Pacific Herring biological samples processed in 2026 in the Haida Gwaii major stock assessment region (SAR). Each sample is approximately 100 fish.

Type	Gear	Use	Number of samples
Test	Seine	Test fishery	4

Table 7. Observed proportion-at-age for Pacific Herring from 2016 to 2026 in the Haida Gwaii major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

Year	Proportion-at-age								
	2	3	4	5	6	7	8	9	10
2016	0.166	0.162	0.170	0.058	0.376	0.044	0.020	0.002	0.002
2017	0.138	0.322	0.100	0.112	0.050	0.200	0.049	0.015	0.014
2018	0.045	0.404	0.242	0.098	0.063	0.072	0.070	0.004	0.002
2019	0.018	0.540	0.312	0.077	0.022	0.018	0.012	0.001	0.000
2020	0.006	0.020	0.751	0.159	0.043	0.013	0.006	0.002	0.001
2021	0.027	0.028	0.039	0.711	0.143	0.036	0.012	0.001	0.003
2022	0.021	0.281	0.031	0.065	0.527	0.072	0.001	0.002	0.000
2023	0.029	0.406	0.215	0.061	0.063	0.166	0.048	0.012	0.000
2024	0.050	0.251	0.314	0.150	0.042	0.045	0.101	0.041	0.006
2025	0.213	0.251	0.159	0.164	0.099	0.023	0.023	0.050	0.016
2026	0.011	0.542	0.215	0.096	0.068	0.034	0.006	0.008	0.020

Table 8. Pacific Herring spawn survey locations, start date, and spawn index in metric tonnes (t) in 2026 in the Haida Gwaii major stock assessment region (SAR). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (NAs).

Statistical Area	Section	Location name	Start date	Spawn index (t)
00	006	Louscoone Inlt E	March 12	705
00	006	Louscoone Inlt W	March 12	156
00	006	Skindaskun Is	March 12	143
02	021	Alder Is	March 29	64
02	021	Alder Is Cr	March 29	626
02	021	Burnaby Str	April 01	610
02	021	De la Beche Inlt	April 02	320
02	021	Huxley Is	March 28	3,101
02	021	Island Bay	April 02	50
02	021	Marco Is	March 29	1,229
02	021	Marshall Inlt	March 28	106
02	021	Saw Rf	March 29	NA
02	021	Scudder Pt	March 29	3,071
02	021	Section Cv	March 29	662
02	021	Section Is	March 29	793
02	021	Sedgwick Bay	March 28	866
02	021	Werner Pt	March 29	NA
02	023	Gray Pt	March 14	NA
02	023	Kitson Pt	March 22	88
02	024	Dana Pass (N End)	April 10	174
02	024	Powrivco Bay	April 11	525
02	024	Selwyn Inlt	April 10	1,019
02	024	Thurston Hrbr	April 10	380
02	025	Poole Inlt	April 02	187
02	025	Rebecca Pt	April 02	333
02	025	Scudder Cr	March 30	570
02	025	Slim Inlt	March 19	304
02	025	Smithe Pt	March 18	624
02	025	Swan Is	March 18	704

Table 9. Summary of Pacific Herring spawn survey data from 2016 to 2026 in the Haida Gwaii major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Units: metres (m), and metric tonnes (t).

Year	Total length (m)	Mean width (m)	Mean number of egg layers	Spawn index (t)
2016	30,345	53	1.1	6,888
2017	31,350	62	0.9	3,015
2018	35,575	44	1.1	4,588
2019	77,965	40	1.4	11,623
2020	47,950	75	2.8	20,423
2021	48,300	78	2.0	18,233
2022	33,250	56	1.0	5,280
2023	25,775	52	1.2	1,583
2024	43,400	45	1.7	11,732
2025	52,925	48	1.9	8,947
2026	72,395	49	2.0	17,424

8 Figures

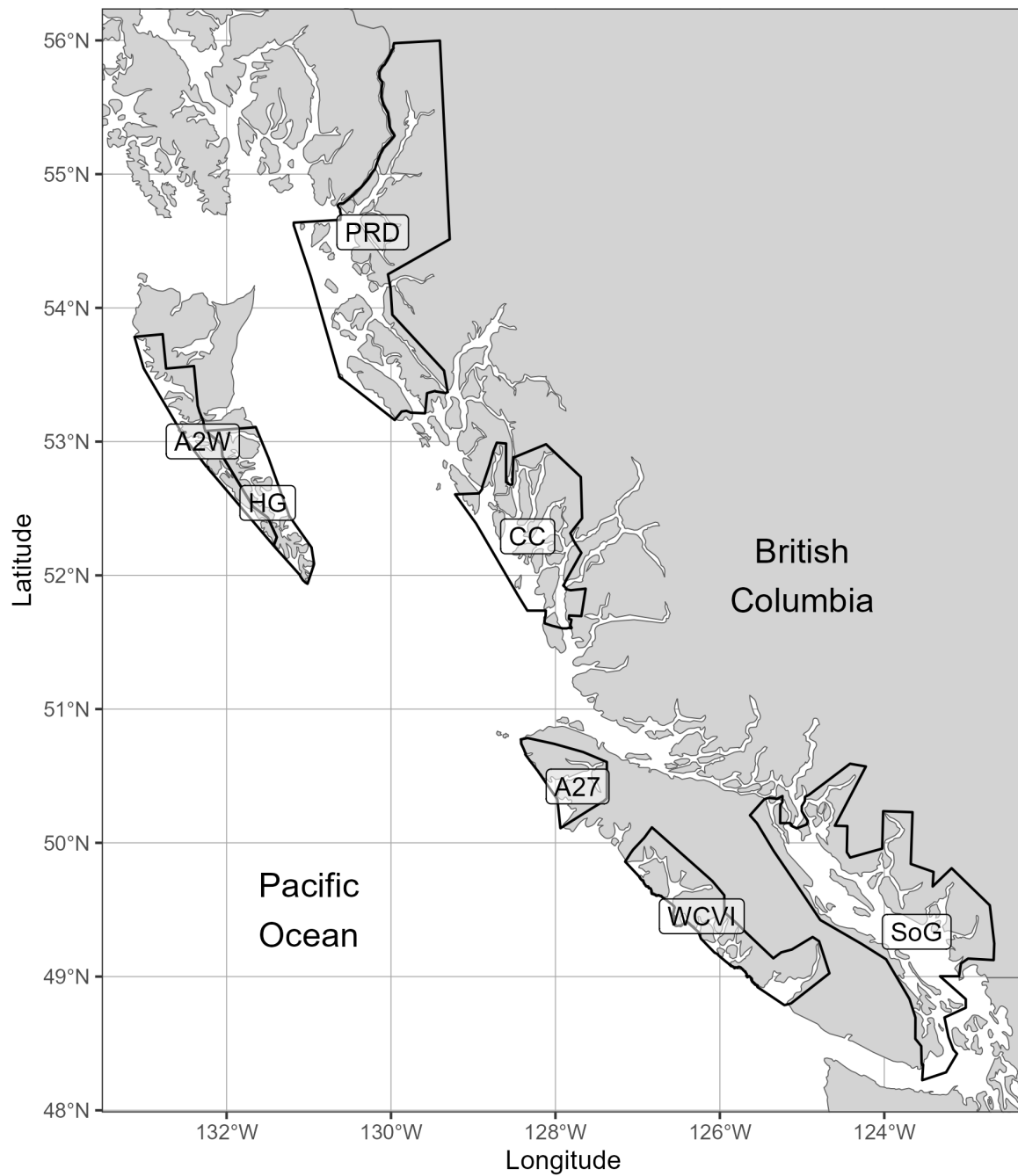


Figure 1. Boundaries for the Pacific Herring stock assessment regions (SARs) in British Columbia. There are 5 major SARs: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), Strait of Georgia (SoG), and West Coast of Vancouver Island (WCVI). There are 2 minor SARs: Area 27 (A27) and Area 2 West (A2W). Units: kilometres (km).

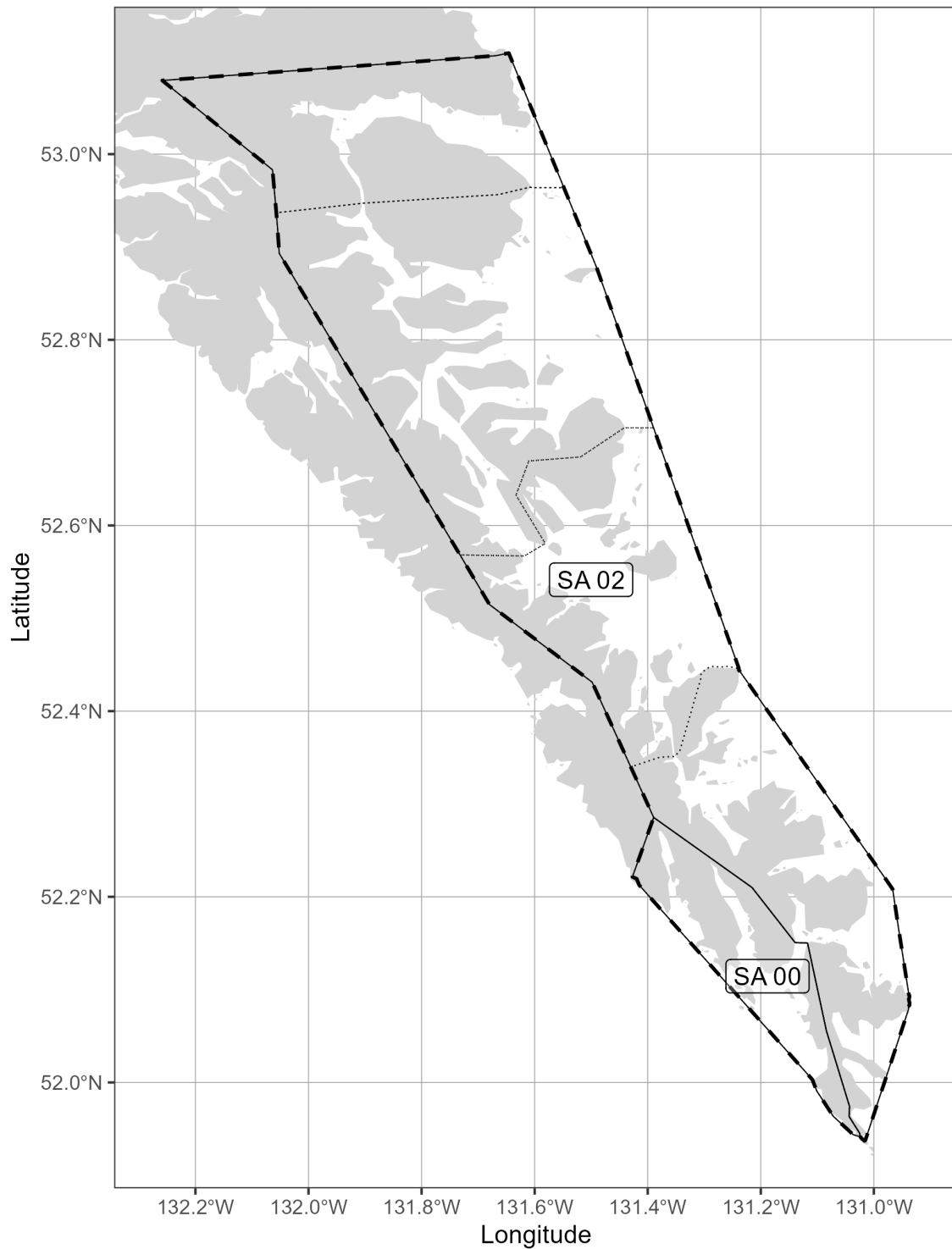


Figure 2. Boundaries for the Haida Gwaii major stock assessment region (SAR; thick dashed lines), associated Statistical Areas (SA; thin solid lines), and associated Sections (thin dotted lines). Units: kilometres (km).

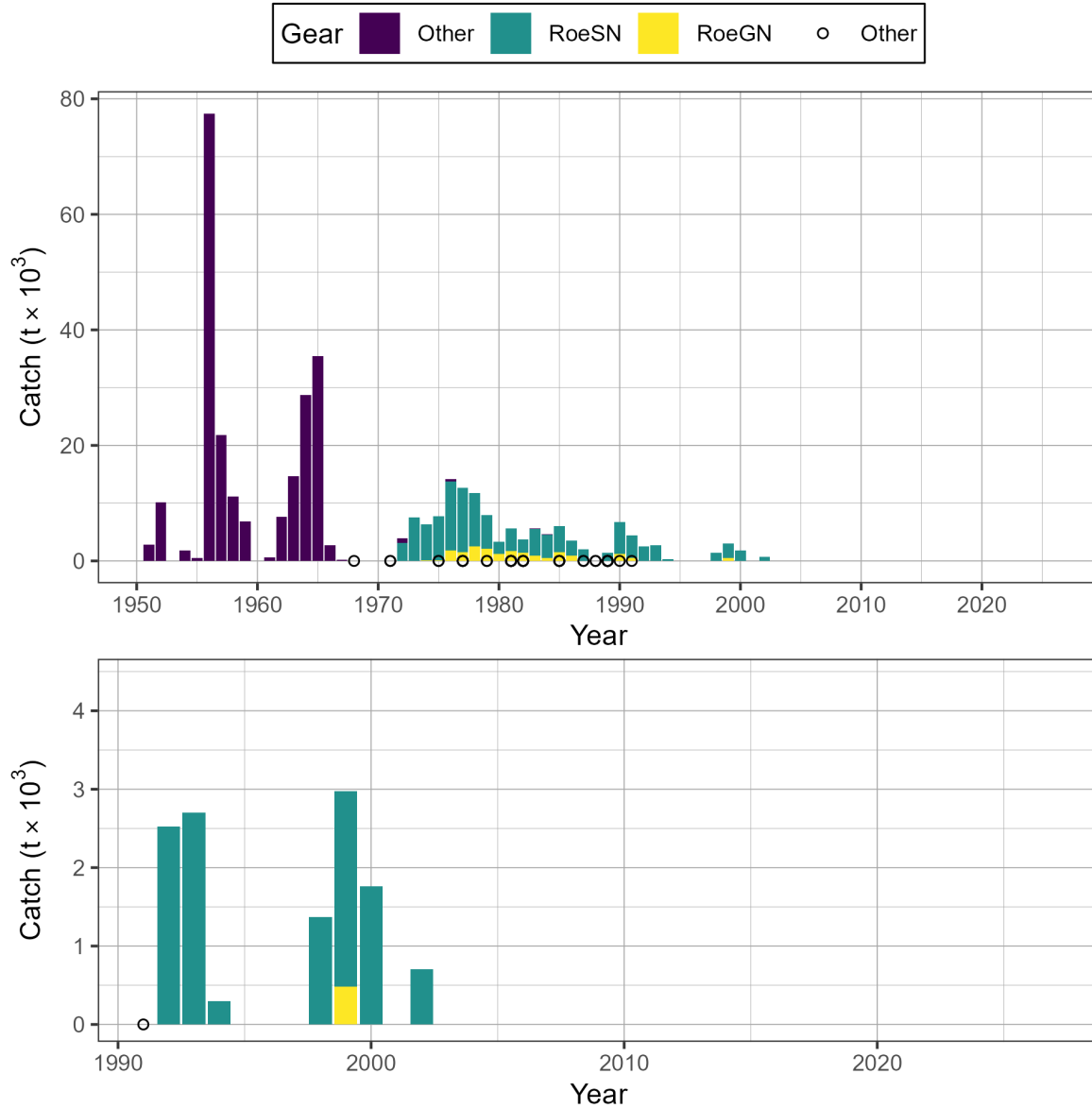


Figure 3. Time series of total landed catch in thousands of metric tonnes ($t \times 10^3$) of Pacific Herring by gear type from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). Legend: ‘Other’ represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; ‘RoeSN’ represents the roe seine fishery; and ‘RoeGN’ represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Bottom panel shows catch since 1991 in more detail. Note: symbols indicate years in which catch by gear type (i.e., Other, RoeSN, RoeGN) is withheld due to privacy concerns.

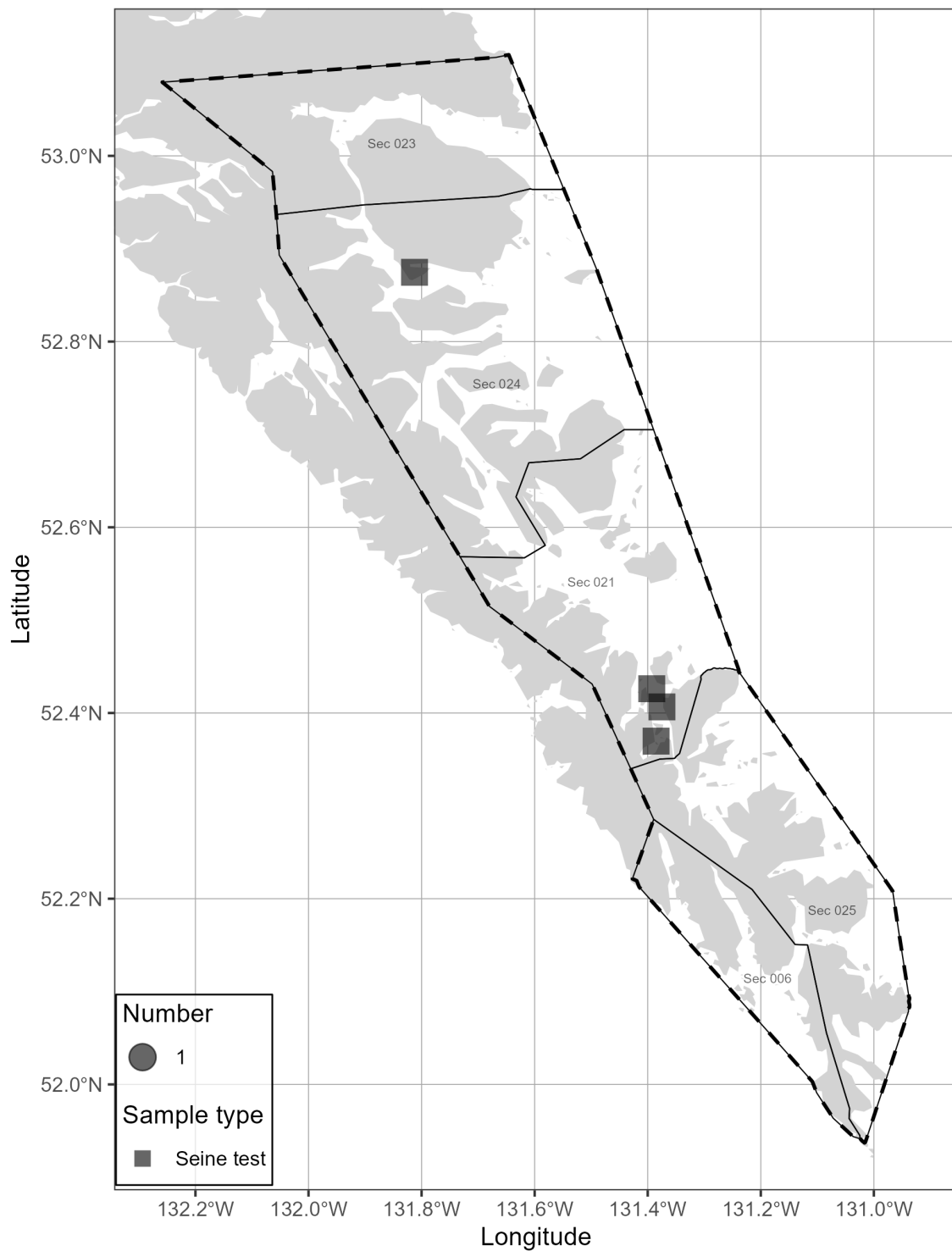


Figure 4. Location and type of Pacific Herring biological samples collected in 2026 in the Haida Gwaii major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). Units: kilometres (km).

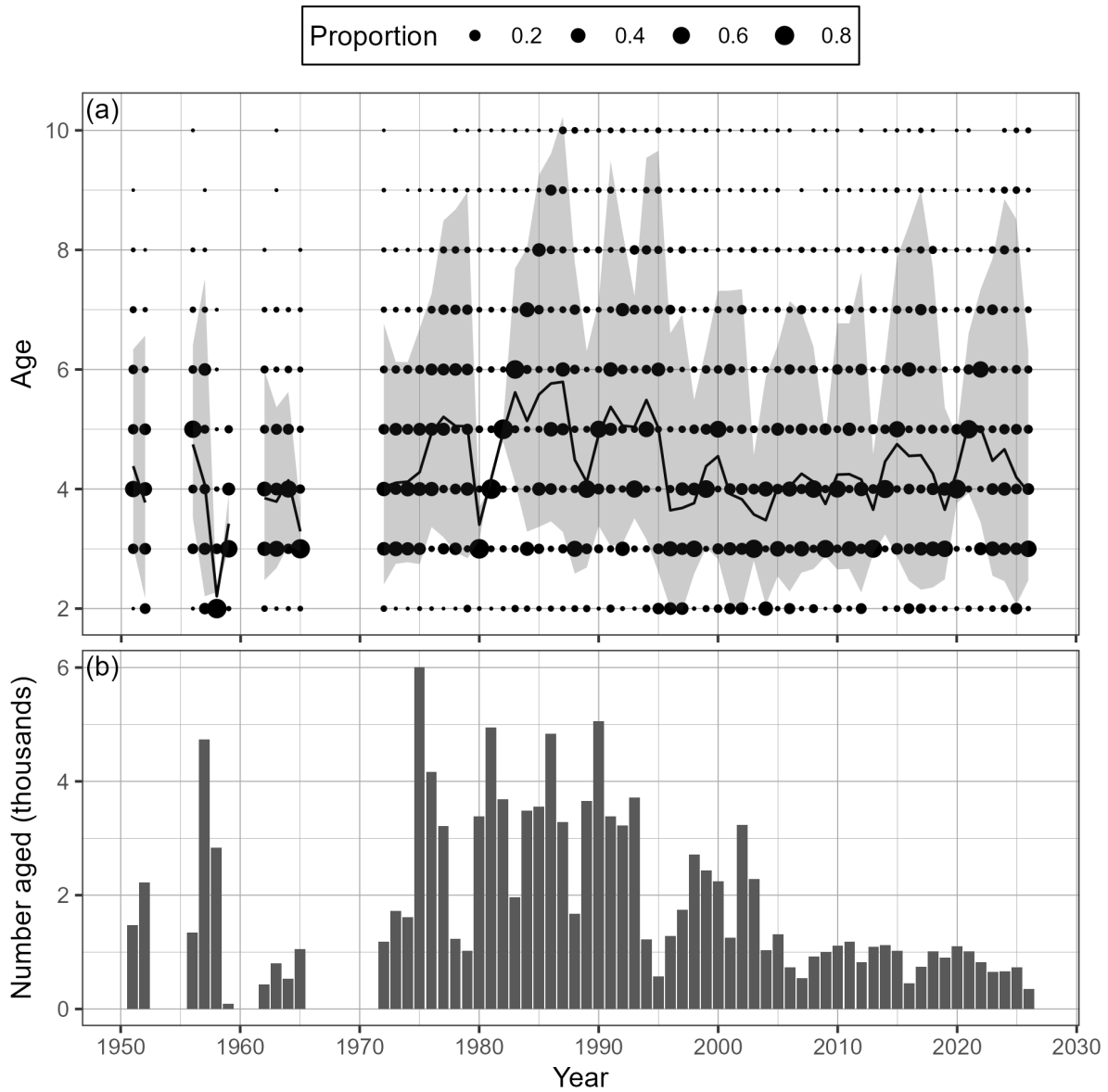


Figure 5. Time series of observed proportion-at-age (a) and number aged in thousands (b) of Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). The black line is the mean age, and the shaded area is the approximate 90% distribution. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

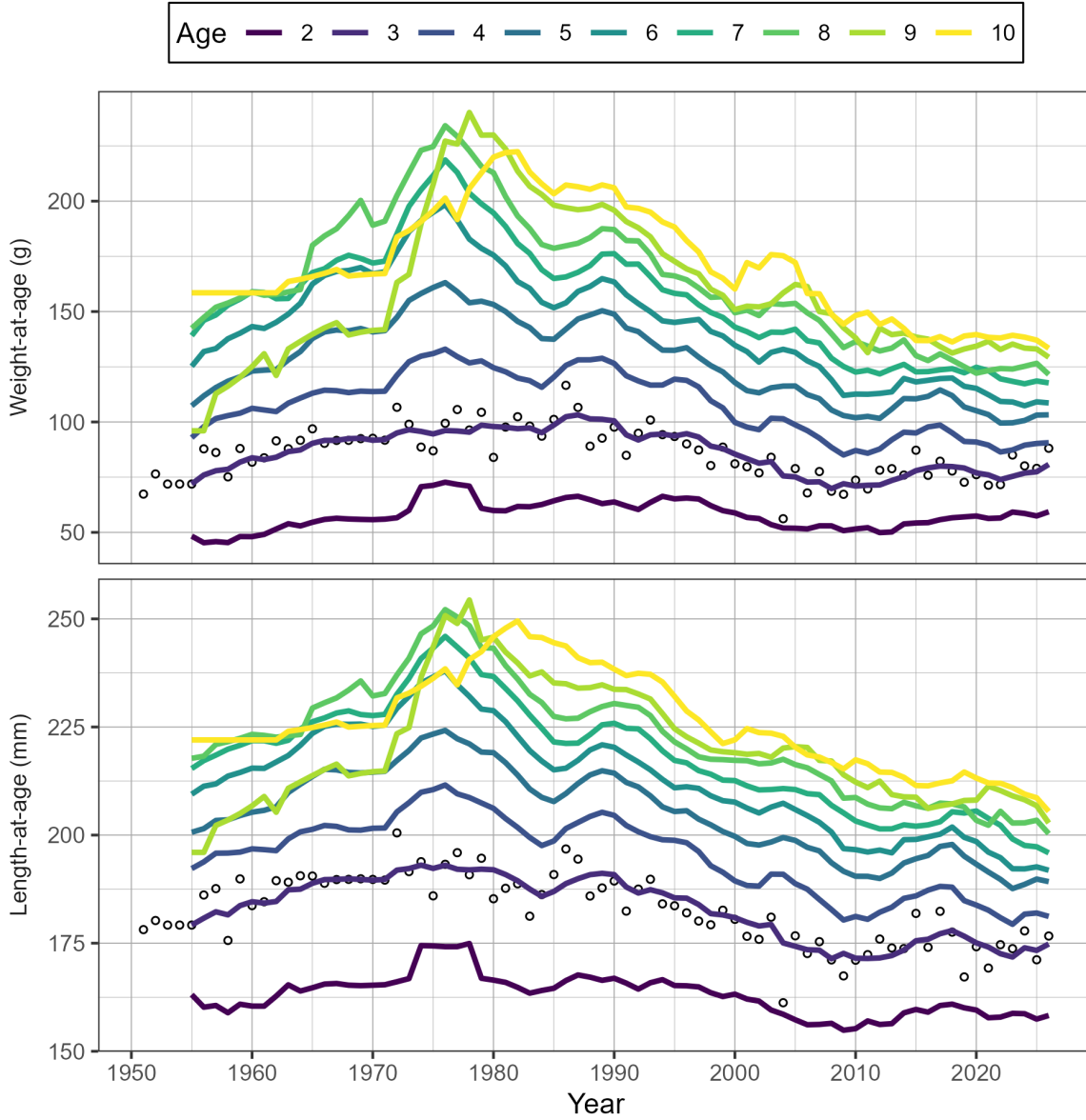


Figure 6. Time series of weight-at-age in grams (g) and length-at-age in millimetres (mm) for age-3 (circles) and 5-year running mean weight- and length-at-age (lines) for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). Missing weight- and length-at-age values (i.e., years with no biological samples) are imputed using one of two methods: missing values at the beginning of the time series are imputed by extending the first non-missing value backwards; other missing values are imputed as the mean of the previous 5 years. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

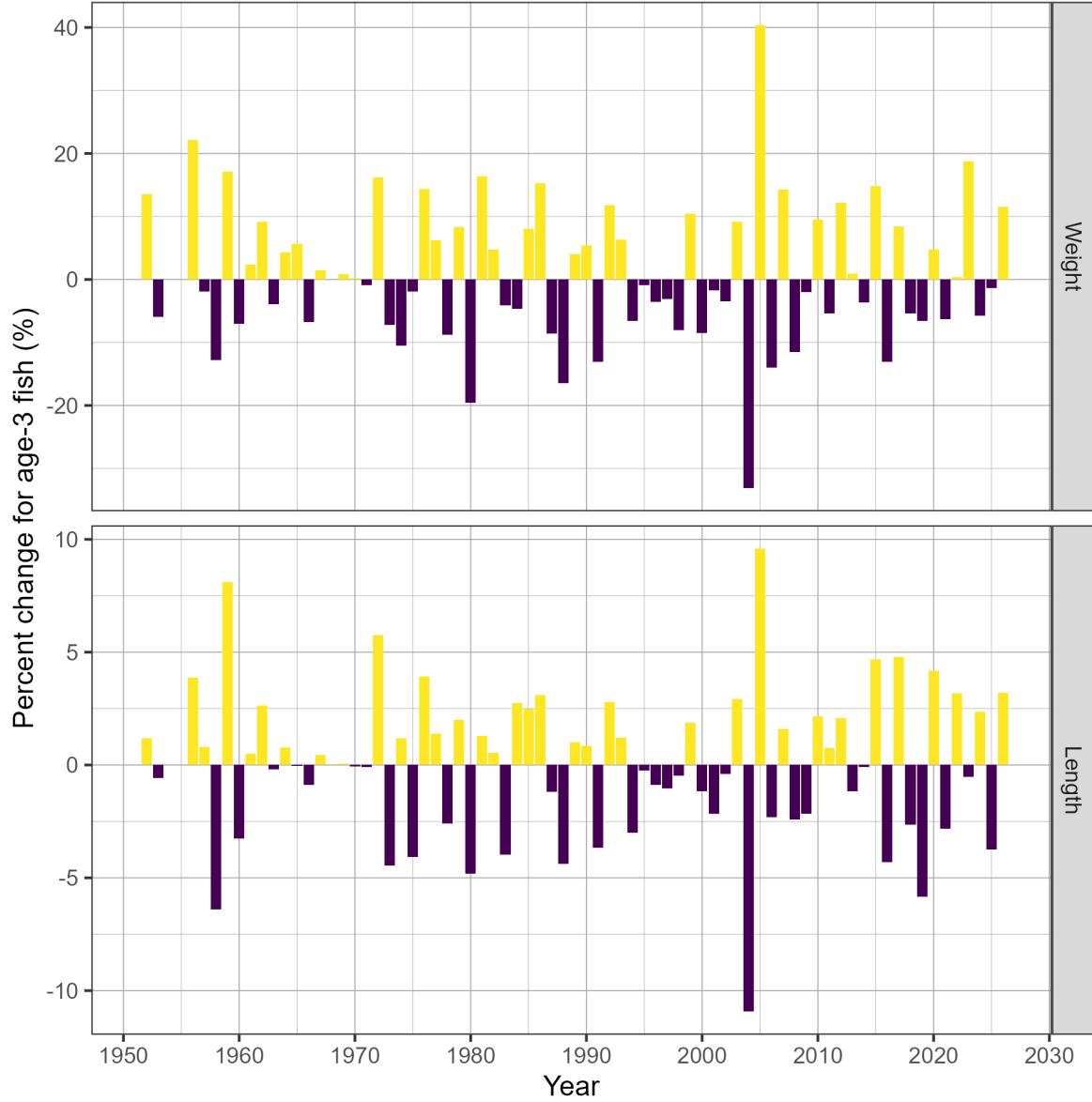


Figure 7. Time series of percent change (%) in weight and length for age-3 fish for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-3 fish, respectively, in year t . Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

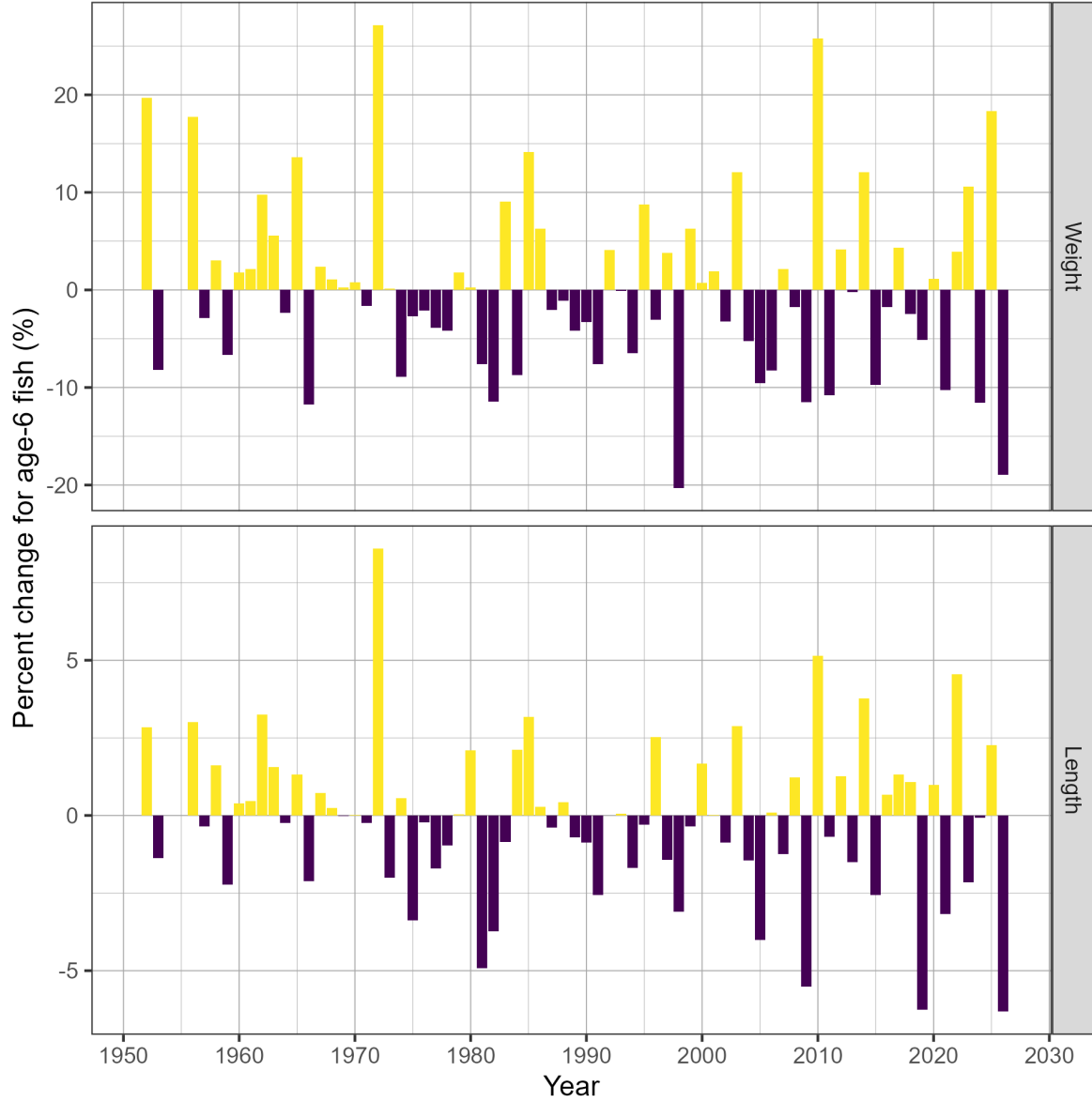


Figure 8. Time series of percent change (%) in weight and length for age-6 fish for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-6 fish, respectively, in year t . Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

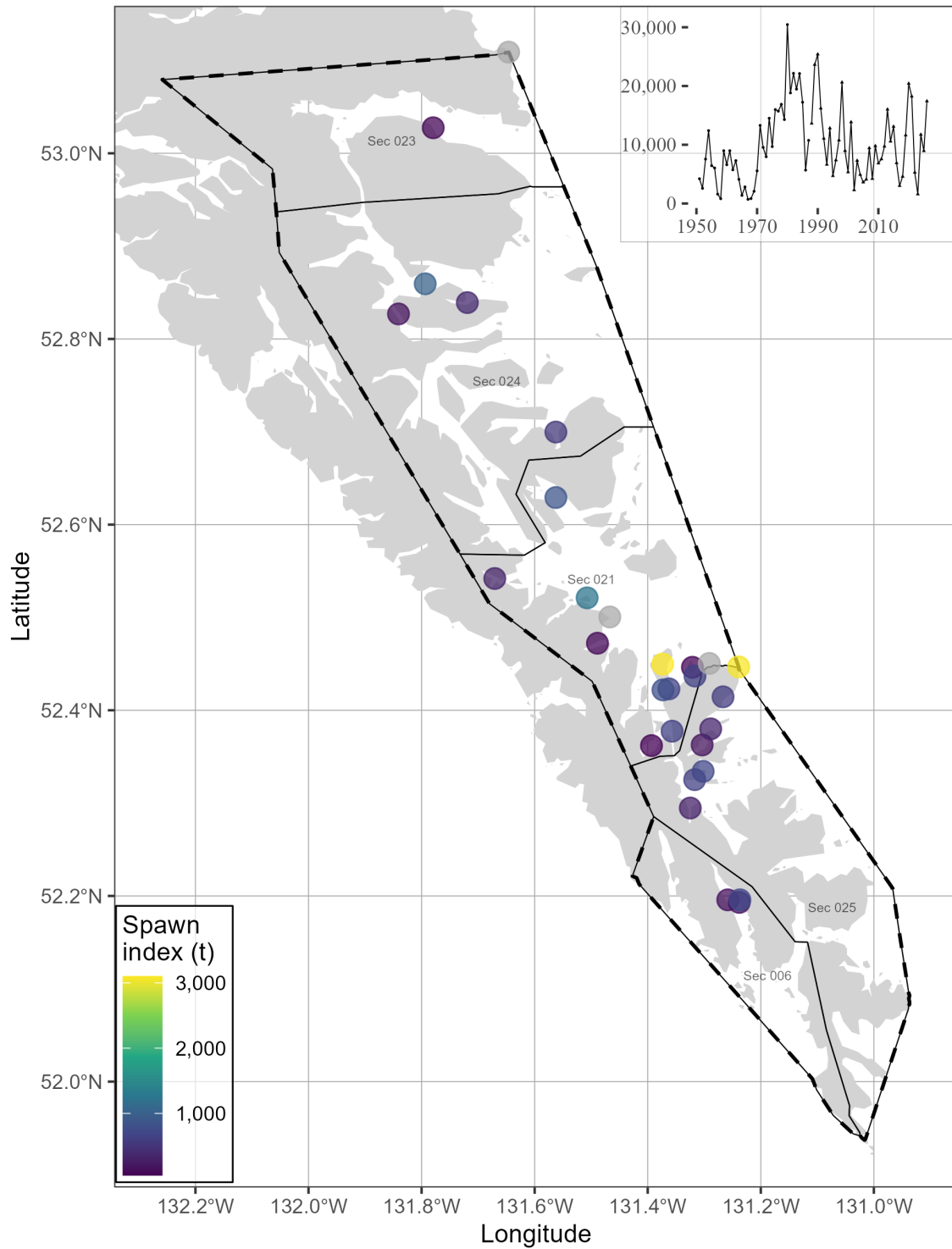


Figure 9. Pacific Herring spawn survey locations, and spawn index in metric tonnes (t) in 2026 in the Haida Gwaii major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km).

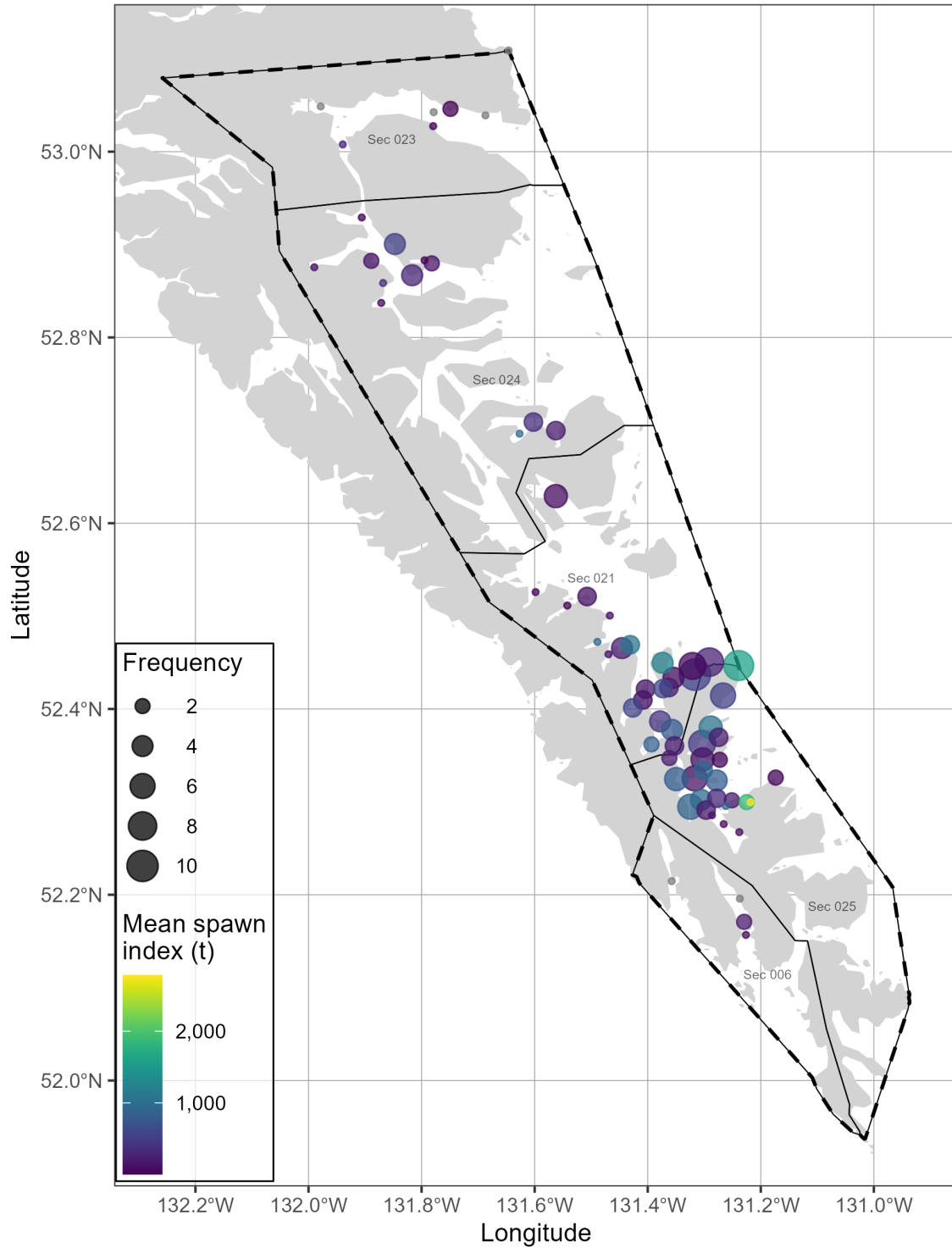


Figure 10. Pacific Herring spawn survey locations, mean spawn index in metric tonnes (t), and spawn frequency from 2016 to 2025 in the Haida Gwaii major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Units: kilometres (km).

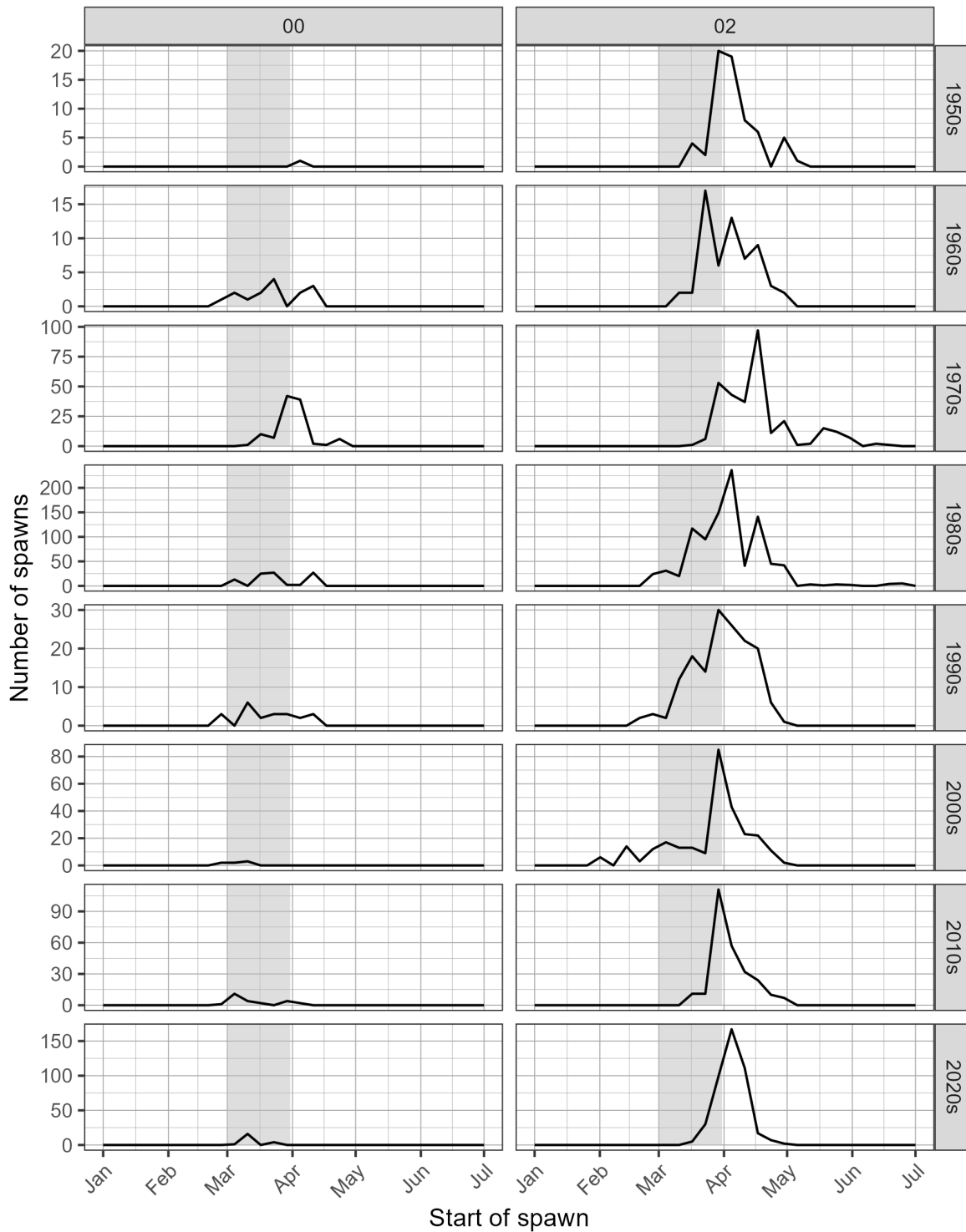


Figure 11. Pacific Herring spawn start date by decade and Statistical Area. Grey shaded regions indicate March 1st to 31st. Note that spawn size and intensity varies; therefore the number of spawns is not directly proportional to spawn extent or biomass.

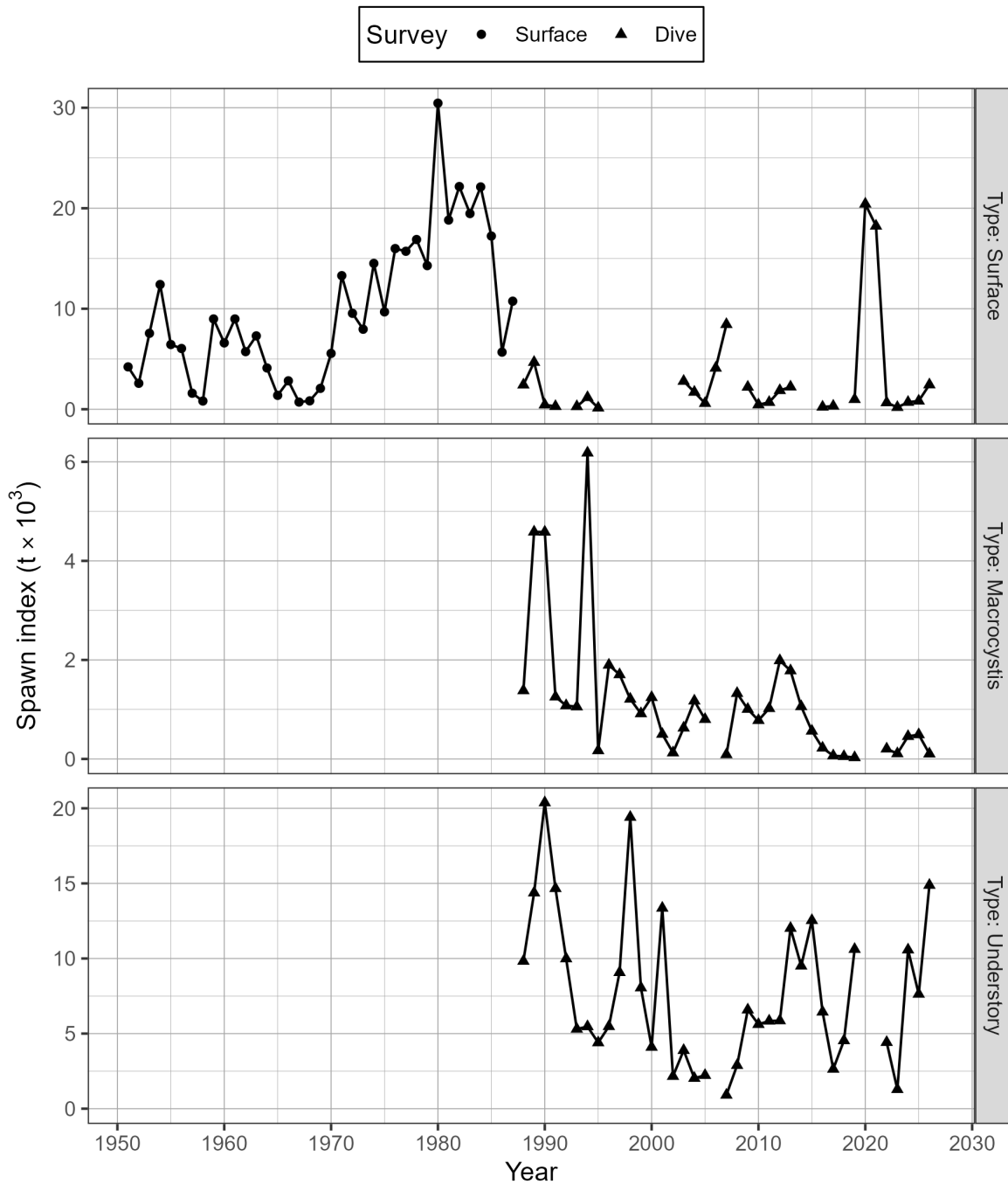


Figure 12. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) by type for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). There are three types of spawn survey observations: observations of spawn taken from the surface usually at low tide, underwater observations of spawn on giant kelp, *Macrocystis* (*Macrocystis* spp.), and underwater observations of spawn on other types of algae and the substrate, which we refer to as ‘understory.’ The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). See Figure 13 for the spawn survey method.

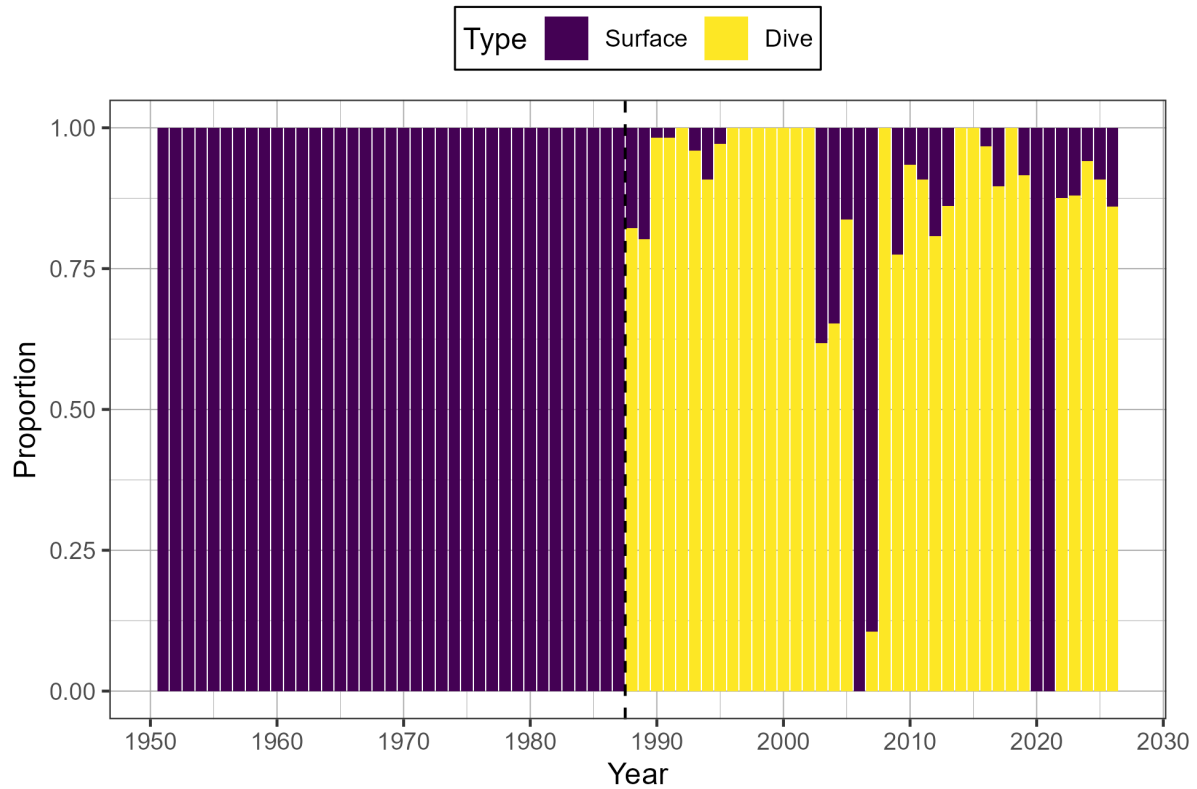


Figure 13. Time series of proportion of spawn index by surface and dive surveys for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

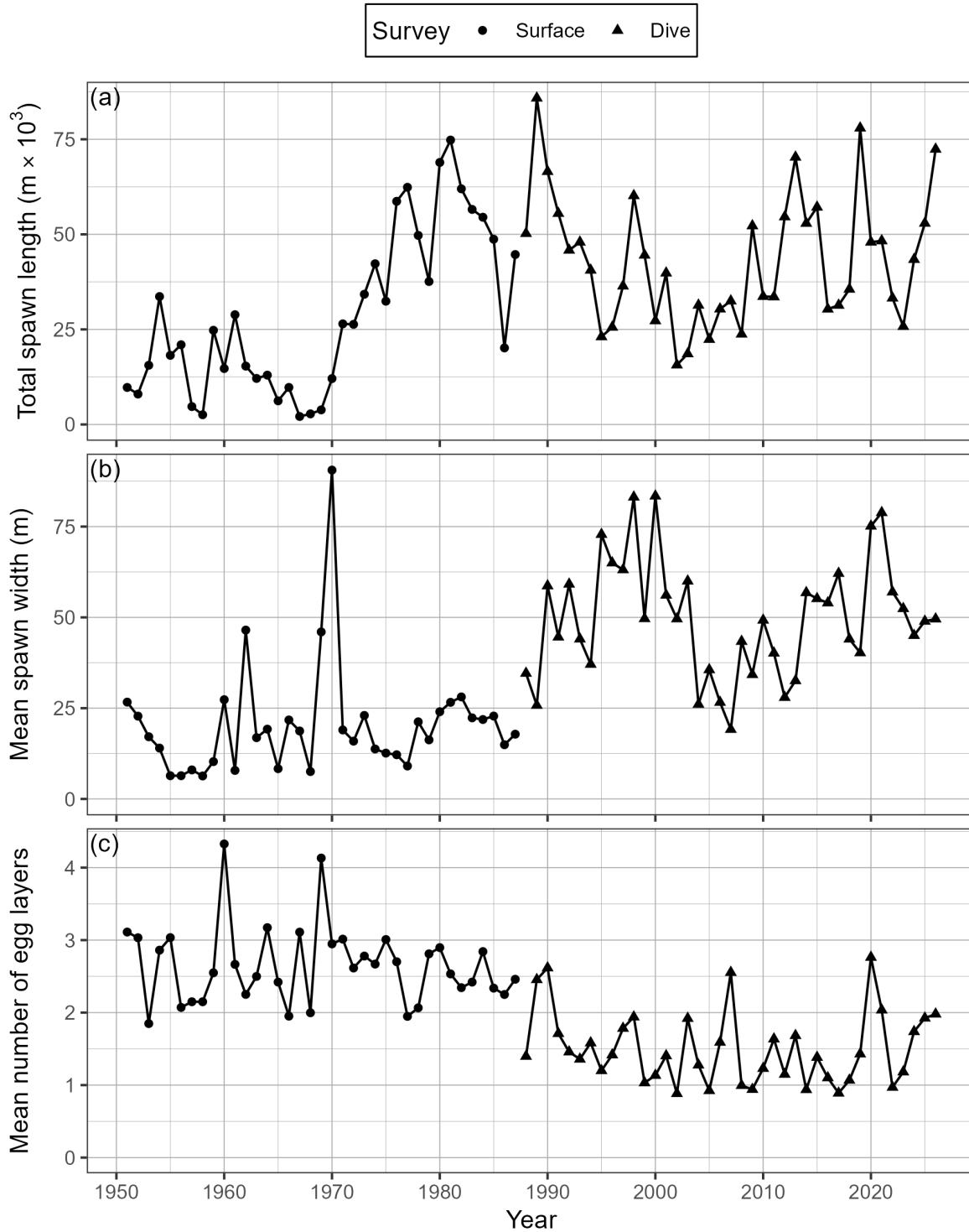


Figure 14. Time series of total spawn length in thousands of metres ($m \times 10^3$; panel a), mean spawn width in metres (b), and mean number of egg layers (c) for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

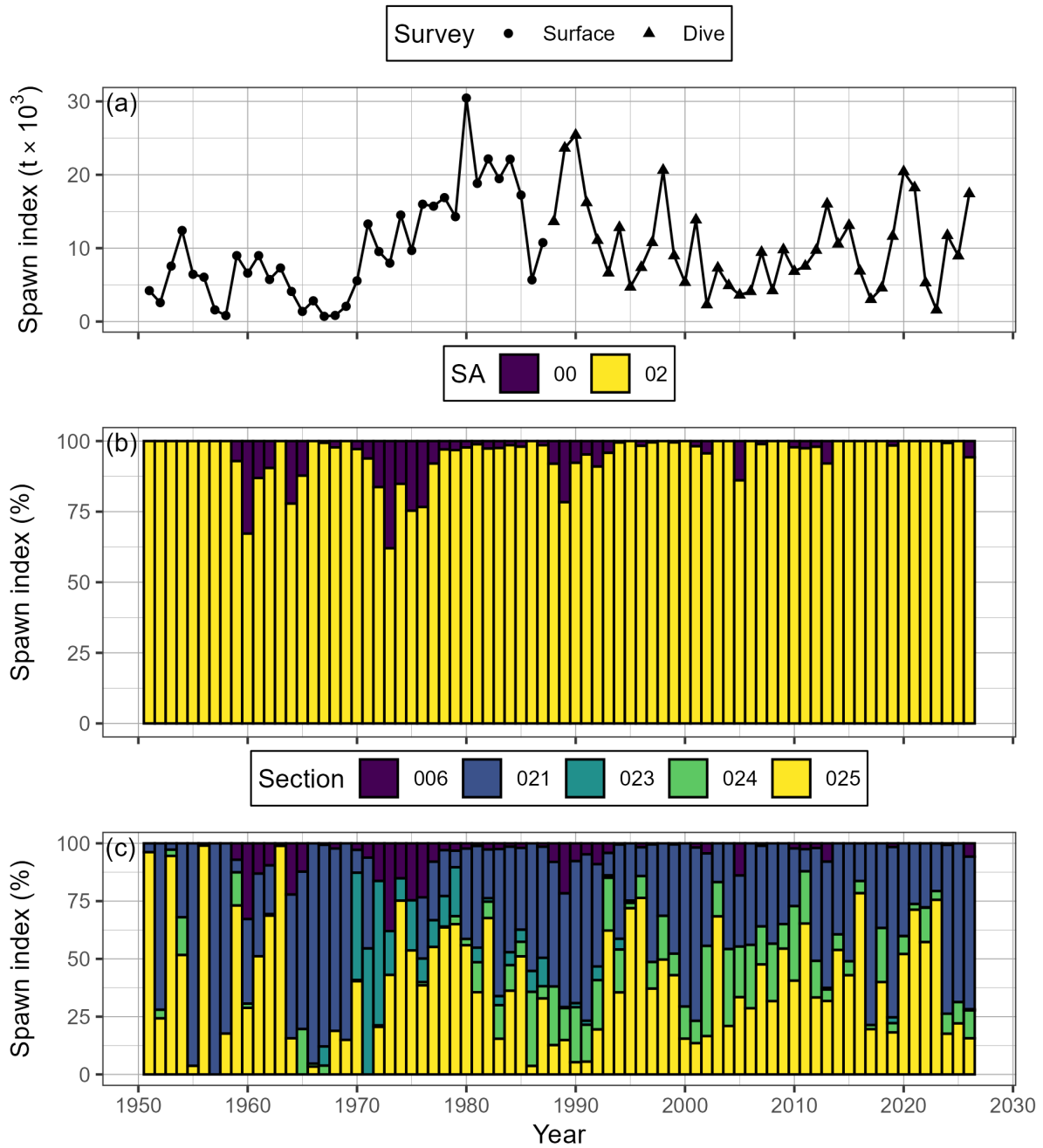


Figure 15. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR; panel a), as well as percent contributed by Statistical Area (SA), and Section (b, & c, respectively). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). Note that spawn surveys in the dive survey period (1988 to 2026) are a combination of surface and dive surveys (Figures 12 and 13). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q .

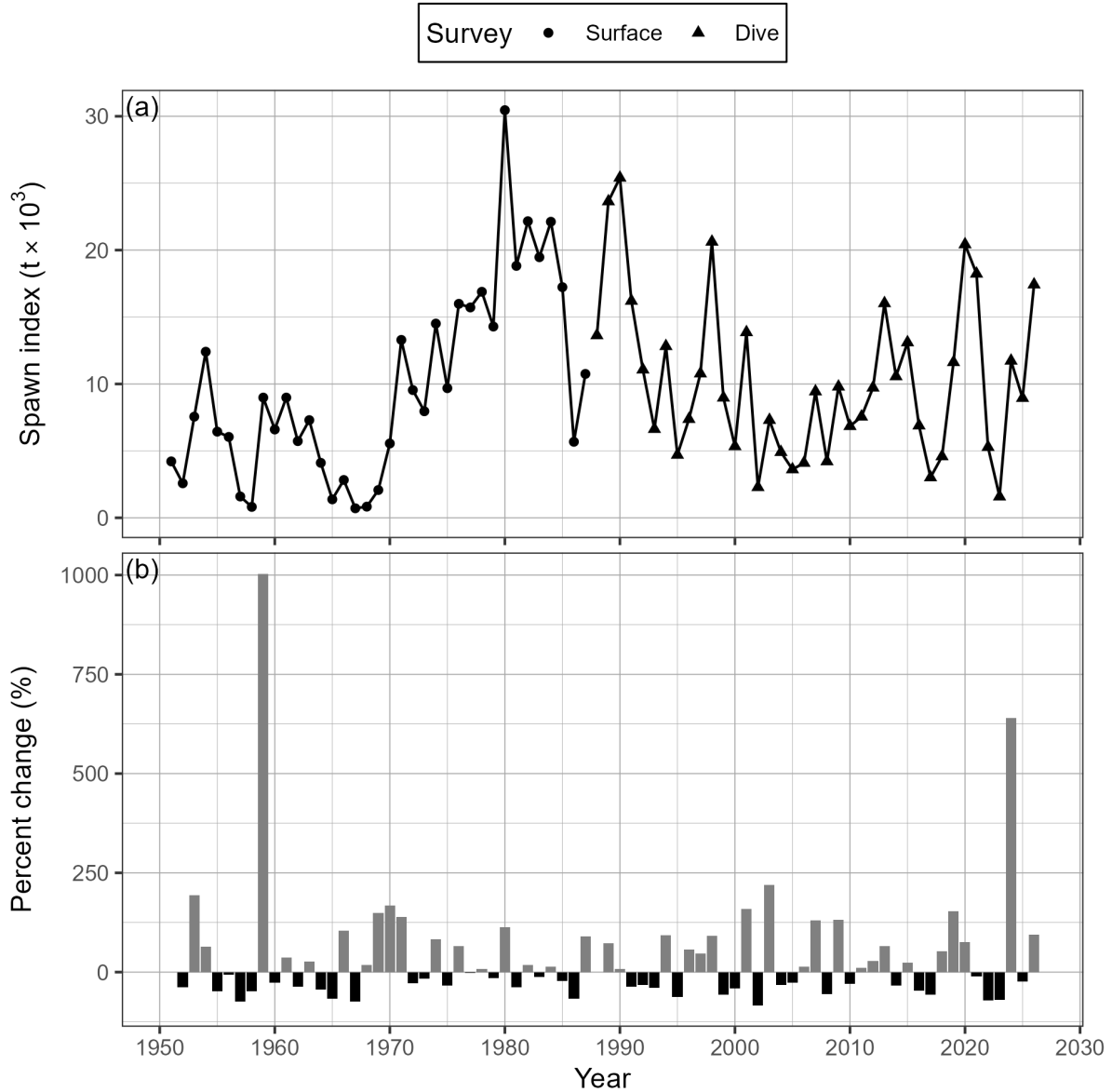


Figure 16. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR; panel a), and percent change (b). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the spawn index in year t . The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Note that spawn surveys in the dive survey period (1988 to 2026) are a combination of surface and dive surveys (Figures 12 and 13).

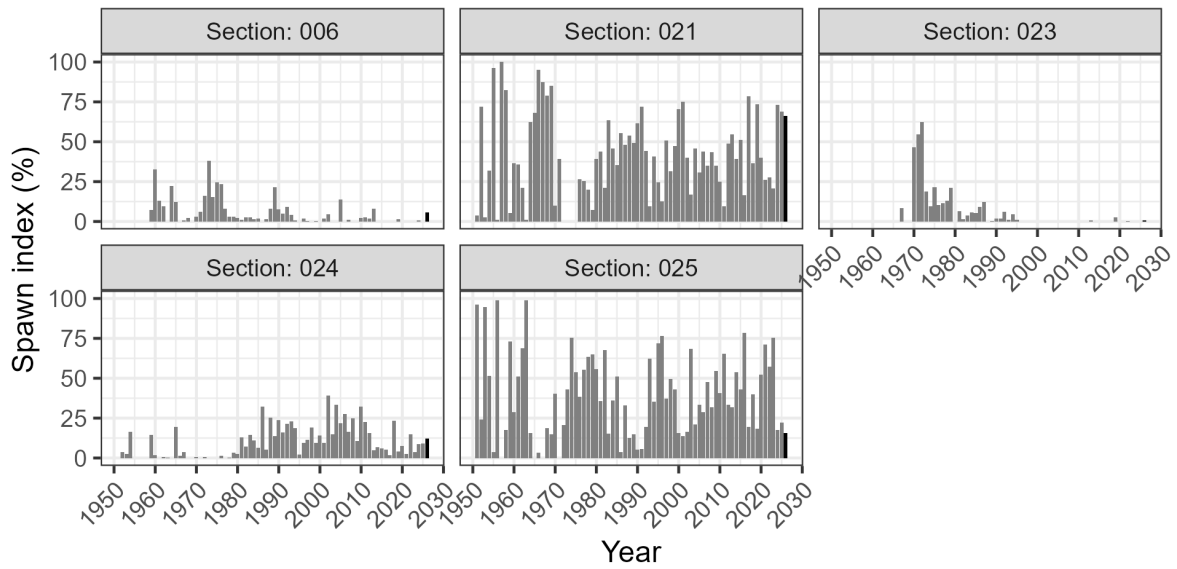


Figure 17. Time series of percent of spawn index by Section for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). The year 2026 has a darker bar to facilitate interpretation. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q .

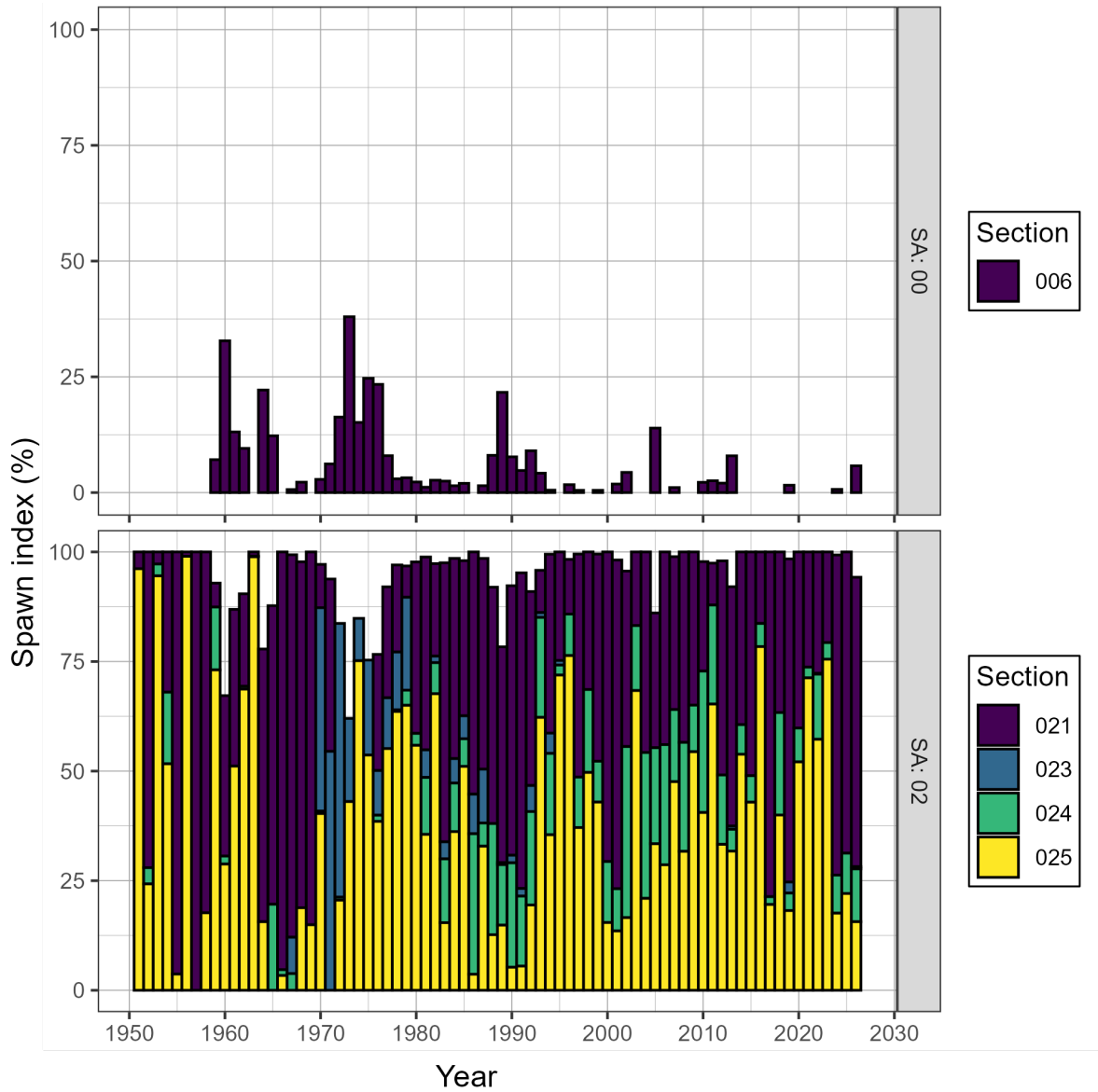


Figure 18. Time series of percent of spawn index by Statistical Area (SA) and Section for Pacific Herring from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q .

Figure 19. Animation of Pacific Herring spawn index in metric tonnes (t) by Location from 1951 to 2026 in the Haida Gwaii major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km). View the animation: [download the report](#), open with Adobe, enable Java, and click “play”.